Skill Builder Pack 2 - Cleaning Data in Excel Using Power Query

Purpose

After data has been imported into Excel, it is rarely ready to use immediately.
Data exported from systems often contains issues that can affect analysis. These may include:

- Incorrect formatting
- Blank rows or columns
- Extra spaces in text
- Inconsistent column names
- Incorrect data types
- Duplicate records

If these issues are not addressed, they can lead to incorrect analysis or misleading results.

For example, a column containing dates stored as text cannot be used correctly in time-based analysis, and names containing extra spaces may appear as separate entries even though they refer to the same person.

For this reason, data analysts spend a significant amount of time cleaning and preparing datasets before analysis begins.

Excel provides a powerful tool called Power Query that allows datasets to be cleaned in a structured and repeatable way.

This skill builder focuses on learning how to prepare datasets so that they are consistent, accurate and suitable for analysis.

Learning Outcomes

By the end of this skill builder you should be able to:

- Open imported datasets in the Power Query Editor
- Identify common data quality issues
- Correct data types
- Rename columns clearly
- Remove blank or unnecessary rows
- Clean text fields using trimming tools
- Ensure datasets use consistent identifiers

These skills are part of the data preparation stage of the analysis process.

Starting Point

You should already have an Excel workbook containing the datasets imported during Skill Builder Pack 1.

These datasets include:

- Attendance records
- Student details
- Engagement scores
- Withdrawal records

Each dataset currently exists as a separate table in Excel.

In this skill builder you will open each dataset in Power Query and prepare it for later analysis.

Opening Power Query

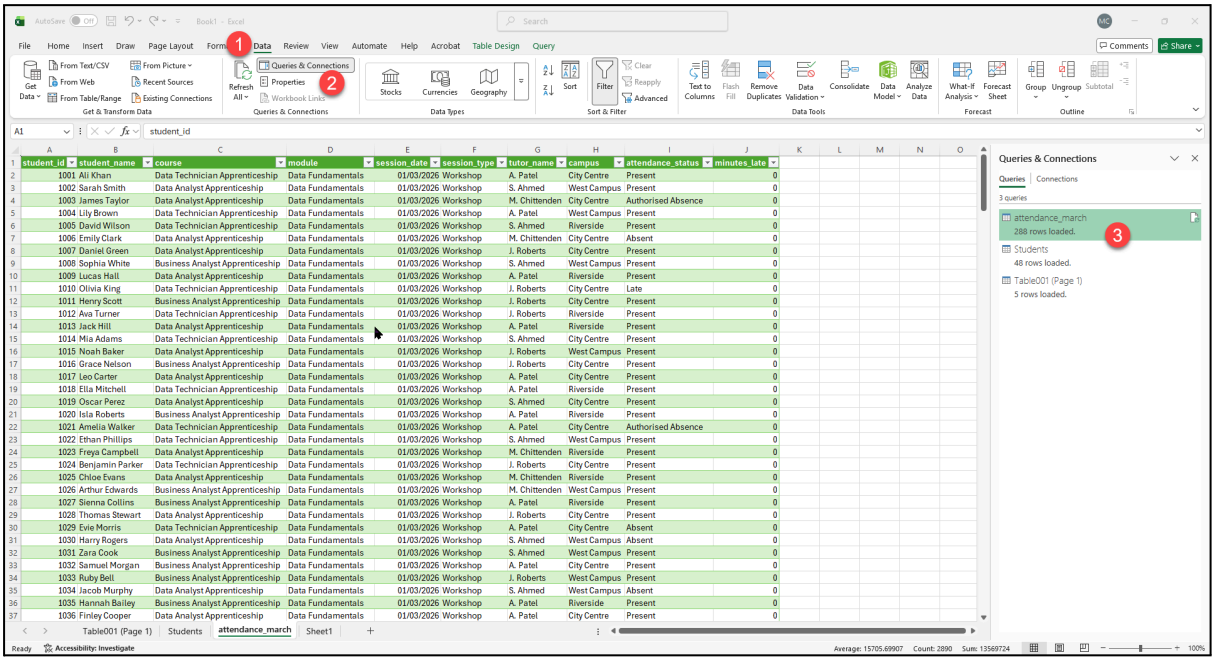
To clean a dataset, you must first open it in Power Query.

Follow these steps:

1. Select the Data tab in Excel

2. Open **Queries & Connections**

3. Locate the dataset you want to edit



4. Double-click the query name

This opens the Power Query Editor.

The editor allows you to perform data cleaning tasks such as:

- Changing formats
- Removing rows
- Editing column names
- Transforming text

Every change you make is recorded as a step. This means that the same cleaning process can be automatically repeated whenever the data is refreshed.

Skill 1 – Understanding Data Types

Each column in a dataset has a data type, which tells Excel how the values should be interpreted.

Common data types include:

Data Type	Example
Text	Names or categories
Whole Number	Student ID numbers
Decimal Number	Scores or measurements
Date	Session dates

If the data type is incorrect, Excel may not analyse the data properly.

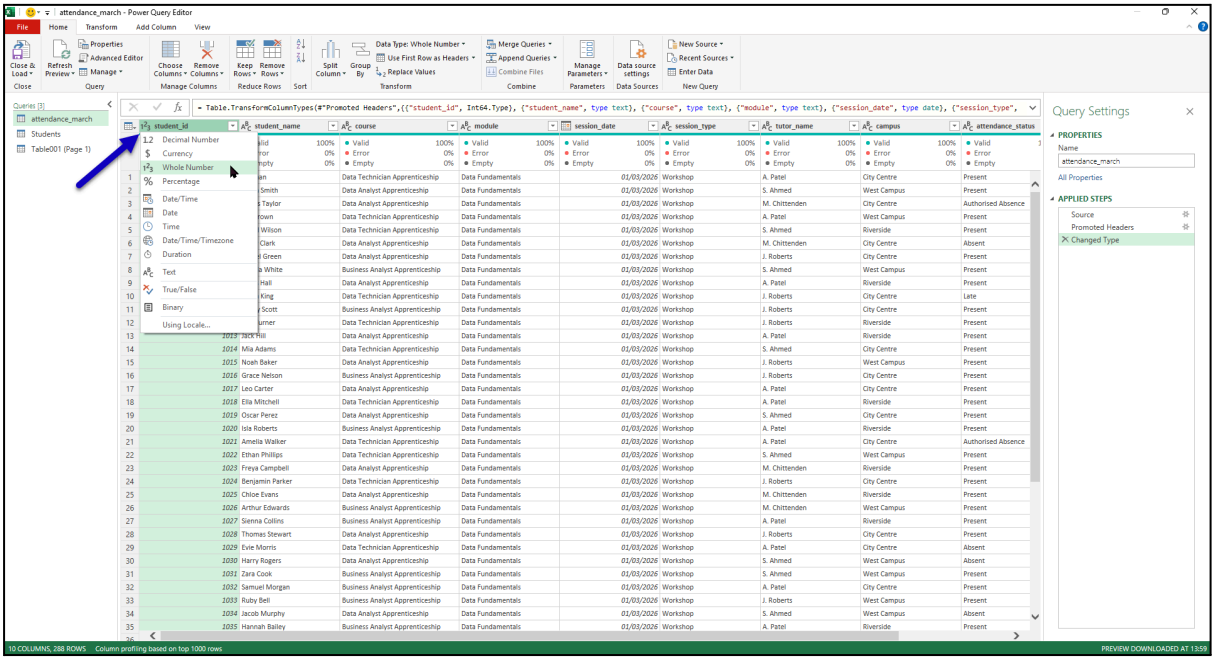
For example:

- Dates stored as text cannot be used in date calculations.
- Numbers stored as text cannot be summed or averaged.

Practice Activity

Open the attendance dataset in Power Query.

Check the following columns by clicking on the icon to the left of the column name:



- student_id
- session_date
- attendance_status

Ensure that:

- student_id is set to Whole Number
- session_date is set to Date
- attendance_status is set to Text

If a column has the wrong type, change it using the data type icon next to the column name.

Skill 2 – Renaming Columns

Column names should be clear and easy to understand.

When datasets come from different systems, column names may be inconsistent or unclear.

For example:

Poor Name	Improved Name
ID	student_id
Date	session_date
Status	attendance_status

Clear naming helps ensure datasets can be understood and combined correctly.

Practice Activity

Review column names in each dataset.

Make sure that:

- Column names are clear
- Similar fields use consistent naming
- Important identifiers (such as student_id) have identical names across datasets

Skill 3 – Removing Blank Rows

Exported datasets often contain blank rows that do not contain any data.

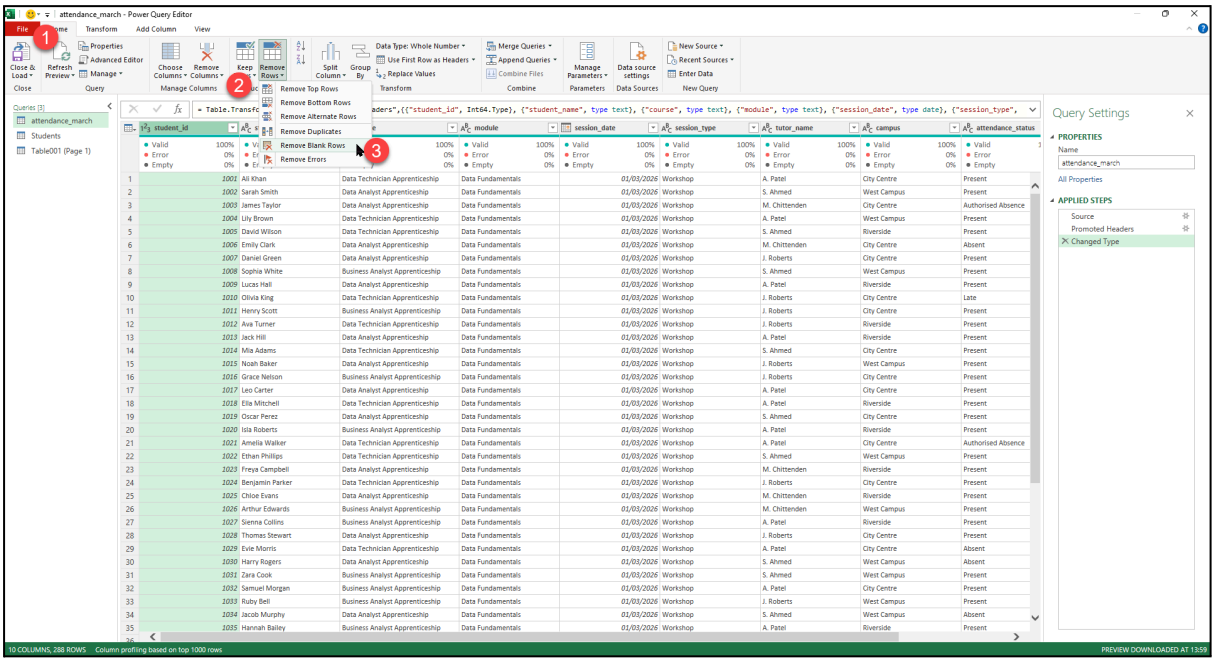
These rows can cause problems during analysis and may affect calculations or charts.

Power Query makes it easy to remove them.

Practice Activity

Open the attendance dataset.

1. Select the Home tab in Power Query
2. Choose Remove Rows
3. Select Remove Blank Rows



Check that only rows containing real records remain.

Skill 4 – Cleaning Text Fields

Text data often contains unwanted spaces.

For example, the following values may appear identical but are treated differently by Excel:

Ali Khan
Ali Khan
Ali Khan

These variations can cause duplicate entries or incorrect grouping.

Power Query provides tools to clean text values.

Practice Activity

Open the **Students** dataset.

Select the following columns:

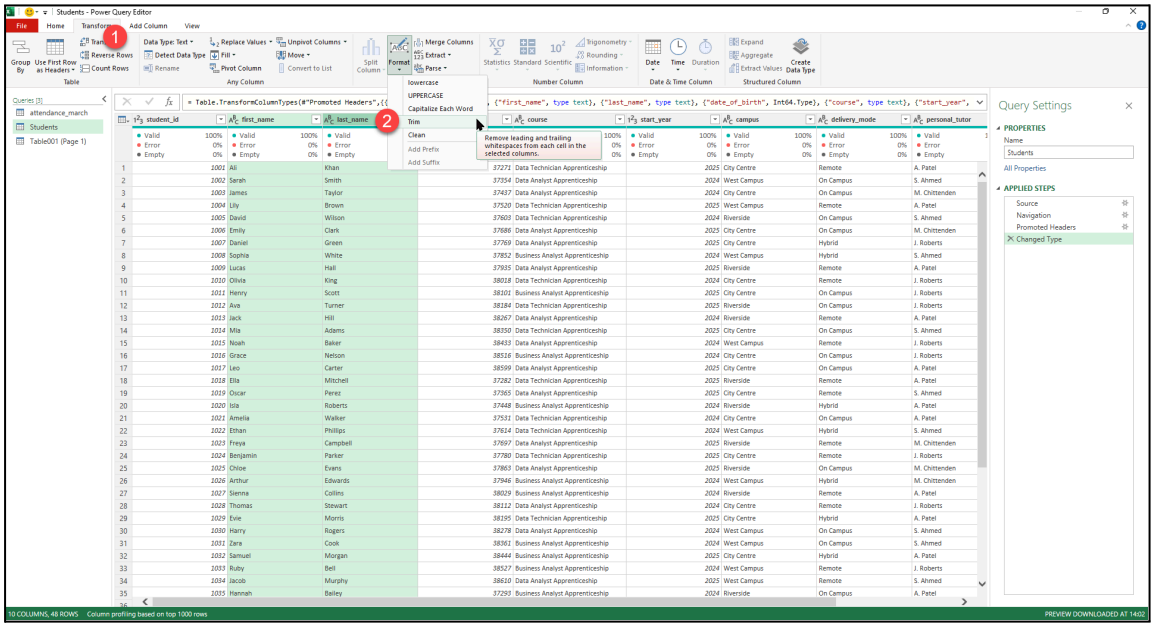
- first_name
- Last_name

If you want, press **SHIFT** to multi-select.

Then:

1. Go to the Transform tab
2. Choose Format → Trim

This removes spaces at the beginning and end of text values.



Skill 5 – Checking Data Consistency

Datasets often need to be combined later using a shared identifier.

In this case, the key identifier is:

student_id

If this field is inconsistent across datasets, the data cannot be joined correctly.

Practice Activity

Review the student_id column in all datasets.

Check that:

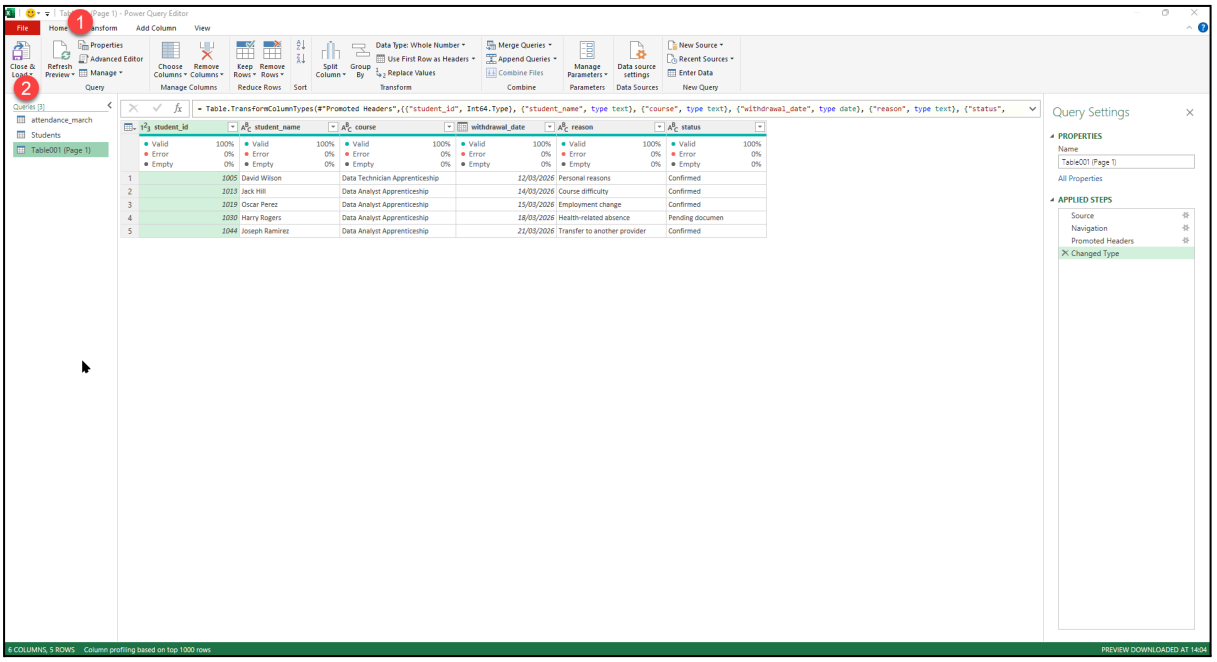
- The column name is identical
- The data type is the same
- Values appear consistent

This step ensures the datasets can later be combined successfully.

Loading Cleaned Data Back into Excel

Once you have completed the cleaning steps:

1. Select Home
2. Click Close & Load



The cleaned datasets will be loaded back into Excel.

All transformation steps are saved.

This means that if the source data is updated later, Excel can simply refresh the queries and repeat the cleaning automatically.

Key Concept

Cleaning data is a critical stage of data analysis.

Analysts rarely begin with perfect data. Instead, the typical workflow is:

Import → Clean → Combine → Analyse → Visualise

In this skill builder you have focused on the Clean stage.

Preparing datasets carefully helps ensure that any later analysis is accurate and reliable.